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ORIGINAL PAPER

A phenomenological-based model for a spray drying tower

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Abstract
This work describes a detailed phenomenological-based semi-physical model (PBSM) for a co-current spray drying tower used for maltodextrin drying. The model is useful for process optimization and control. This model differs from previously published models by using an alternative numerical methodology, reducing the loss of information. The droplets hydrodynamics, particles, and air are modeled along with heat and mass transfer, combining the movement of the macroscopic phase and transfer phenomena at a lower scale (droplet and particle). The model is a tool to analyze drying mechanisms and processes dynamically, considering the axial changes but using a simple algorithm structure. The advantages of the model are demonstrated by comparing its results with published data. The modeled dynamics are compared to similar system experimental data, obtaining reasonable and expected outcomes. The steady-state obtained is compared with reported data from a computational fluid dynamics (CFD) model for a co-current spray drying tower.

Keywords Co-current spray tower · Spray drying · Phenomenological-based model · Maltodextrin

Introduction

Spray drying is an operation unit commonly used by the chemical, food, and pharmaceutical industries to produce and finish several everyday products. It provides product stability and extends shelf life by reducing the moisture content to levels that no longer support microbial growth (Masters 2002). The mathematical modeling of this operation unit represents a promising tool for developing new designs, troubleshooting and improving existing units, analyzing conditions, optimizing and controlling its operation.

Despite the extensive use of spray drying, a predictive analysis of the phenomena has been challenging given the complexity of the interactions between disperse and continuous phases (Ali et al. 2020). The variable velocities of both phases and the highly coupled mass and heat transfer make the phenomena complex to model. Furthermore, during the drying, the airflow pattern affects the particle trajectories, making the process dependent on the position inside the dryer. As a result, the drying history of the droplets varies among them, generating variations to the final product.

Many authors have researched spray drying mathematically to improve the understanding of the process. Some of these studies focus on the drying mechanisms and study single droplet behavior (Farid 2003; Mezhericher et al. 2008b; Handscomb and Kraft 2010; Schutyser et al. 2012). These studies help explain drying phenomena but do not explain the drying behavior in industrial systems and conditions. Indeed, gas phases are not generally specified as a modeled subsystem. Other authors focus on detailed models on computational fluid dynamics (CFD) based on different mass transfer equations or specific drying mechanisms (Mezhericher et al. 2012; Schmitz-Schug et al. 2016; Kieviet 1997; Jaskulski et al. 2017; Mezhericher et al. 2008a).

Tools such as CFD in modeling have allowed a comprehensive analysis of the phenomena improving current designs, the understanding, and the scaling-up of these units. It is known that the computational requirements of the CFD models are high, representing a disadvantage to the control and applications in real-time. However, many efforts have been made to develop a dynamic model for decision-making during online operations as process control. Still, Lisboa

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et al. (2018) mention the existence of a gap in the construction of a simple, accurate, robust, and easily implemented model to estimate drying conditions in industrial spray dryers. Current work tries to close this gap.

Other models focused on representing the steady-state and dynamic of industrial spray dryers separately (Ali et al. 2014; Julklang and Golman 2013; Birchal et al. 2006; Shabde and Hoo 2006; Cotabarren et al. 2018; Sivarajalingam 2009; Petersen 2016; Negiz et al. 1995). In these works, different numerical methodologies are used to solve the mathematical model, whose complexity promotes the consideration of many assumptions and several simplifications. For example, the drying has been simplified using a plug flow reactor (PFR) approach, whereby the dryer is a counter-current tower (Ali et al. 2014), to a continuous stirred tank (CSTR) approach, whereby the dryer is a mixed flow spray dryer (Sivarajalingam 2009; Petersen 2016).

Assumptions and simplifications applied to these models imply a trade with reality and phenomena explanation but allow for solving them more efficiently. Similarly, another approach for simplifications is to not consider the dynamic behaviour of the continuous phase by using equations for the air that do not depend on their accumulation term related to time but only on terms depending on space, such as the PFR simplification, which disregards the accumulation term (Ali et al. 2014). These simplifications help resolve and give general information, mostly about the final product and gas stream, but affect the model's capacity to give specific information, such as drying history and dynamic behaviour of the gas phase during the drying. Birchal et al implemented a numerical methodology which considered the spray dryer in the model prescinding from assumptions as CSTR systems, considering the dynamic behaviour of both phases, and achieving extracting data for droplet history.

The model presented in this work approaches the spray drying modelling similarly to Birchal et al. However, the numerical methodology presented differs in the calculus of the interactions of both phases. Instead of considering the dispersed phase as a stream that flows through compartments, this methodology calculates the effect of groups of droplets crossing the dryer one by one and then superposes the responses to obtain the continuous response of the system, making this a method that allows following in detail the behaviour of certain groups of droplets when they experience disturbances or different conditions in the process. Additionally, this methodology allowed applying simple numerical methods such as the Euler, implementing spatial constraints without sacrificing extracting specific information with simplifications such as drying history and keeping the dynamic behaviour of both phases giving space for control systems.

This study aims to present a whole model for spray drying and a new numerical methodology for solving the model, considering axial changes of the droplet and air phases without neglecting their accumulation terms. As these terms appear explicitly in the balance equations, the model can be used directly for real-time simulations as process control with a straightforward implementation. This numerical methodology divides the dispersed phase into droplet batches and considers the continuous phase a sum of partitions.

The development of the model was based on a method for obtaining Phenomenological Based Semiphysical Models (PBSM) (Lema-Perez et al. 2019). The method is applied to obtain a model that describes the dynamic behavior of the two phases inside a spray drying tower. Furthermore, the simulation predicted an expected behavior of the fluid phase and the droplets that are transformed into powder. This simulation is compared to a similar industrial-scale unit operation.

This work is organized as follows. "Review of droplet drying" section reviews the spray drying process theory, emphasizing the single droplet drying stages. "Model pre-construction" section presents the model pre-construction as a stage prior to the model construction described in "Model construction" section. "Model simulation and contrasting" section presents the model simulation and compares data taken from previous works to illustrate the model response. Finally, conclusions are provided in "Conclusion" section.

Review of droplet drying

The fundamentals of spray drying are based on two parts: atomization and the drying mechanism. Regarding the former, the proposed model uses a pressure atomizer, which discharges liquid through a small aperture, allowing the pressure energy to transform into kinetic energy and produce the droplets. In this way, the atomizer increases the area of the liquid to achieve high drying rates. The mathematical representation of this system gives the droplet velocity, size, and discharge angle, which are relevant initial conditions for the drying process.

The mentioned conditions, the temperatures, and the velocity of the phases affect the drying mechanism. Indeed, depending on them, different product morphologies may be obtained. However, those conditions are suitable for rapid drying and producing hollow spherical particles in the proposed model. This drying mechanism consists of four stages: (i) heating/cooling up to the wet-bulb temperature, (ii) evaporating at a constant rate, (iii) the crust formation when the critical moisture content is reached, and (iv) the limited mass transfer drying. Figure 1 illustrates these stages for a single droplet.

The first stage, heating or cooling to the wet-bulb temperature, begins after atomization and formation of the droplets.

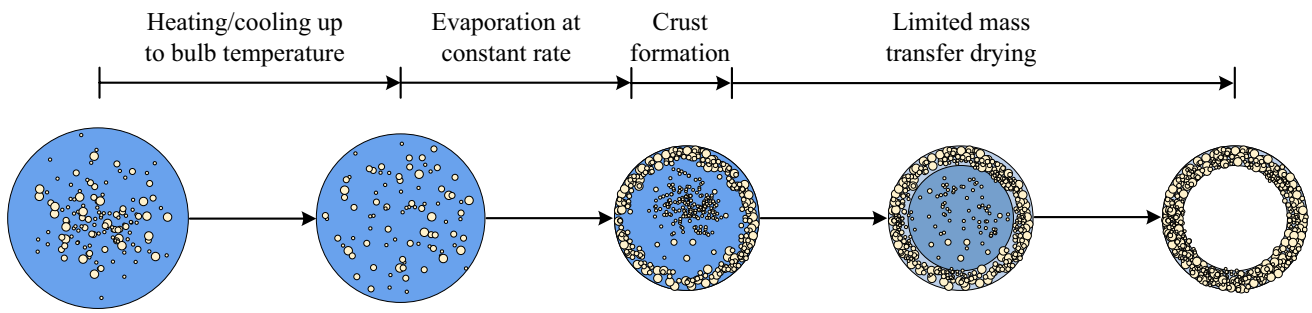


Fig. 1 Drying stages of small droplets during a rapid drying

In the case of an inlet liquid temperature below the wet-bulb temperature, sensible heat is exchanged, and then the mass flux from droplet to air increases from its initial value. In contrast, if the inlet temperature is greater than the wet bulb, a decrease in the mass flux occurs.

Once the wet-bulb temperature is reached, the heat transferred from the air causes water evaporation at a constant rate. In this stage, the water evaporation rate reaches its maximum value. Flux variations occur when the temperature that the droplets are exposed to changes or the decrease of droplet diameter is considerable. It is important to note that the transfer of volatile compounds in this stage occurs on the droplet surface until a crust is formed.

As the droplet diameter decreases due to evaporation, maltodextrin is accumulated as a crust over the droplet. Two main reasons could explain this behavior. The first is the rapid diameter decrease, which could be faster than maltodextrin diffusion toward the droplet center. The second is the friction caused by water molecules moving rapidly toward the droplet edge dragging the maltodextrin molecules with them, or obstructing their movement to the center.

Once the crust has formed, the particles do not contain water on their surface. This means that the mass flux decreases substantially due to the mass transfer through the porous crust. In this stage, heat transfer from the air heats the crust, which then transfers heat to the wet nucleus and porous surface causing evaporation and mass transfer. During drying, the maltodextrin agglomerates along the inner border of the crust. This process is called "bubble nucleation" (Ali et al. 2014). When the solid reaches the equilibrium moisture content, the drying process stops.

Model pre-construction

The modeling pre-construction section is the first part of the mentioned modeling methodology (Lema-Perez et al. 2019). The used PBSM methodology does not have new concepts. It proposes a clear order step to step and a simple guide to junior modellers. This section states a process description, verbal and graphically, along with a modeling hypothesis and its complementary assumptions, and the level of detail of the model. Finally, partitions applied over the process are specified and explained.

Process description and model objective

Co-current spray drying operation begins when there is contact between the continuous and the disperse phases. The former consists of air with a low humidity value, and the latter consists of a volatile compound and the compound to dry, in this case, water, and maltodextrin, respectively. However, there are additional pieces of equipment to contact both phases properly, i.e., the air is heated and filtered before entering the spray dryer through a disperser in the ceiling, and the liquid phase is pumped and enters through a pressure nozzle generating droplets. Then, the produced droplets contact the air, and the phase conditions promote the drying. Finally, particles reach the bottom of the tower, and a valve separates them from the air, while the air with the finest particles is directed to a cyclone.

In this process, atomization is crucial because several initial conditions depend on it. The modeled atomizer is classified as a simplex nozzle or a pressure-swirl type. It enters 0.2 m into the tower and generates the droplets in that position. This atomizer includes three main parts, the atomizer body, the swirl chamber, and an orifice (d_0) of 0.711 mm. During the atomization, the liquid enters the body and then passes through the swirl chamber tangentially. The tangential slots generate the swirling motion of the solution and direct it to the orifice. Due to this movement, the solution leaves the atomizer in a conical sheet that expands and disrupts firstly into ligaments and then droplets. This process is illustrated in Fig. 2.

The current model aims to dynamically represent the size, temperature, and humidity changes of the droplets and solid particles as they descend through the tower (axial direction). For this purpose, it was necessary to track the trajectory of the droplets. A PBSM was built to describe the phenomena

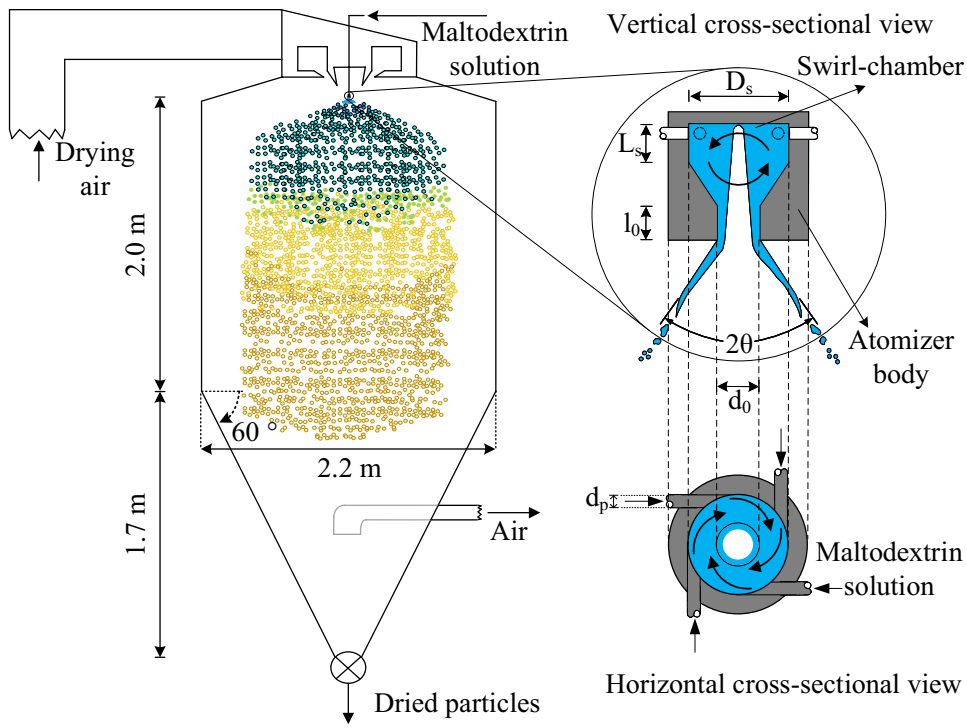


Fig. 2 Schematic view of a co-current spray drying and its simplex swirl atomizer. Adapted from Lefebvre and McDonell (2017)

taking place in a spray under physical changes and operational conditions of the process.

Modeling hypothesis and level of detail

The main hypotheses are related to the morphology of droplets and particles during the drying, the conditions at which crusts are formed, and the gas phase, modeled similar to a PFR but including the accumulation term.

In this case, it is assumed that the nozzle will produce perfect spherical droplets. This sphericity is maintained as the droplets fall and when their diameter is reduced before crust formation. As a result of these assumptions and drying conditions, the particles produced in the drying process are hollow spherical particles.

Another hypothesis is related to the crust formation assuming this accurs at the same maltodextrin concentration for all droplets. Specific variations of the critical moisture content, the concentration at which crust formation occurs, are not considered in this work. Some conditions as excessive heat during the drying may generate expansion and break down of particles, affecting the obtained morphology. Nevertheless, the particles are not exposed to boiling conditions after crust formation. Thus, solid crust expansion is not considered. Below are some other assumptions that complement these hypotheses.

- The input streams to the equipment are fed at constant rates.
- The number of droplets in each partition is the same at any time.
- There is no mass transfer of solvents between the phases.
- There is no chemical reaction between the phases or within them.
- The pressure inside the chamber is uniform and constant.
- Air act as an ideal gas.

The drying phenomena are modeled on a macroscopic scale since each partition retains the order of a few centimeters, considering a drop in size from micrometers to millimeters, still at a macroscopic scale.

Process system definition

The walls of the drying tower delimit its content, the air, and the droplets. There are three phases in the process; the droplets are considered a disperse phase with two aggregation states because as they fall, they can be liquid or solid with a liquid core. Nevertheless, the distinction between the droplets and the gas phase is maintained during the whole process. Thus, the disperse and continuous phases are recognized as process systems (PS) and the liquid contained in the pressure nozzle constitutes another system. Consequently, three PS should be established. The PS scheme is shown in Fig. 3 with heat and mass transfer per partition composed of a defined number of droplets. In accordance with these partitions, all droplets are considered one PS and the air is another PS. Each PS considered is defined as follows.

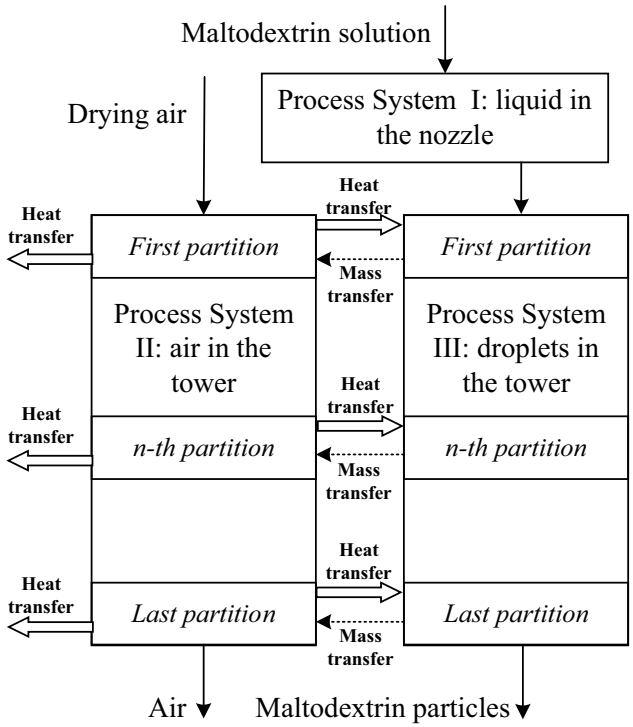


Fig. 3 Process systems taken for modeling the spray drying process

1. Liquid contained in the pressure nozzle is process system I (PS_I), defined as the liquid in the simplex nozzle.
2. Process air is process system II (PS_{II}), defined as the air contained in the drying tower.
3. Process system III (PS_{III}) is defined as the droplets or particles falling through the tower. This process system changes its phase due to mass transfer and subsequent crust formation.

Partitions are defined as mass fractions of phases moving throughout the tower. As a result of this division and movement, each one may be considered a semi-batch in constant displacement and is independent of surrounding partitions of the same phase. In addition, as shown in Fig. 3, each partition has mass transfer only with the other phase. Partition details and balances are stated in the following sections.

In the case of the PS_{III}, a partition of droplets is defined as a group of multiple droplets falling throughout the tower. An air partition is a mass fraction of the air phase at specific heights that also moves along the tower. The velocities between phases are assumed to be vastly different, and the sizes of air partitions are small. Then, a group of droplets fall from one air partition to another in a short period allowing these calculations of discrete events to approach the continuous phenomenon. When a group of droplets stays in an air partition for a limited time, they transfer mass to the air and fall to the next partition. The previous air partition is much slower than discrete groups, so it continues falling and receiving more droplets until it leaves the dryer. Since axial interactions between air partitions are assumed not to be relevant, they can be considered as fed batches receiving groups of droplets and accumulating moisture from them. It should be emphasized that the primary assumption for using this mechanism is the difference in velocities between phases. This assumed mechanism is convenient for simplifying calculus, but using the real conditions of the process. Additionally, as shown in Fig. 3, each partition has mass transfer only with the other phase. Partition details and balances are stated in the following sections.

Model construction

Model basic structure

In this section, the conservation law is applied to each partition, establishing the total mass balance, the component mass balance, and the energy balance. This gives rise to the basic model structure, fulfilling the model objectives established in the process description.

PS_I—liquid contained in the nozzle

Pressure nozzles are usually represented by a set of empirical equations. Specifically, the complexity of the system promotes the use of this type of equations. Empirical equations in this process system do not affect the aim of this work. Thus, this modeling step is skipped, and the empirical equations applied are presented as constitutive equations. However, it is worth mentioning that the importance of this process system resides in the initial drying conditions.

PS_{II}—process air in the spray drying tower

Total mass balance on a drying air partition:

$$\frac{dM_g^j}{dt} = \dot{m}_{evap}^j \tag{1}$$

The drying air or process air inside the tower is divided into independent sections called partitions. An air partition is independent of other partitions, but each one has contact only with the droplets. The air partitions do not mix among themselves; they are independent, considering only the accumulation of moisture coming from the droplets. An air partition in this trajectory only has contact with multiple groups of droplets. In this sense, the variable M_g^j represents the total mass of drying air in a partition j. The term $\dot{m}_{evap}^j$ refers to the water that enters the continuous phase or gas due to evaporation and mass transfer from the droplets. In

addition, as it is not a stagnant phase in the dryer, the way to model the movement of the partitions is detailed in the numerical methodology section.

Component mass balance on a drying air partition:

The results of applying the component mass balance on a partition of the drying air for water (w) is

$$\frac{dM^j_{g,w}}{dt} = \dot{m}^j_{evap} \tag{2}$$

For dry air (a) component, the balance is $\frac{dM^j_{g,a}}{dt} = 0$ as "a" remains constant in each partition.

Energy balance on a drying air partition:

$$\frac{d}{dt}\left(M^j_{g,a}\,\widehat{U}^j_{g,a} + M^j_{g,w}\,\widehat{U}^j_{g,w}\right) = -\left(\dot{Q}^j + \dot{Q}^j_s + \dot{Q}^j_{amb}\right) \tag{3}$$

where $\widehat{U}^j_{g,n}$ is the specific internal energy of compound n of the continuous phase, i.e., air (a) or water (w), $\dot{Q}^j$ the heat transfer from the PS_{II} to the PS_{III}, $\dot{Q}^j_s$ the sensible heat to the evaporated vapors, and $\dot{Q}^j_{amb}$ the heat transfer from the PS_{II} to the tower environment. Besides, the total energy equation $\widehat{U} = C_v\,T$ is applied to each derivative, resulting in

$$\frac{dT^j_g}{dt} = \frac{-\left(\dot{Q}^j + \dot{Q}^j_s + \dot{Q}^j_{amb}\right) - U^j_{g,w}\,\frac{dM^j_{g,w}}{dt}}{M^j_{g,a}\left(T^j_g\,\frac{dC^j_{v\,g,a}}{dT} + C^j_{v\,g,a}\right) + M^j_{g,w}\left(T^j_g\,\frac{dC^j_{v\,g,w}}{dT} + C^j_{v\,g,w}\right)} \tag{4}$$

The term T^j_g is the temperature of the drying air and $C^j_{v\,g,n}$ is the heat capacity at a constant volume of component n of this phase.

PS_{III}—droplets or particles in the tower

Total mass balance on a partition of the disperse phase:

$$\frac{dM^j_p}{dt} = -\dot{m}^j_{evap} \tag{5}$$

A specific number of droplets compose the disperse phase or particles (p). The total mass of droplets in a partition is represented by M^j_p.

Component mass balance on a partition of the disperse phase p:

The results of applying the component mass balance on a partition of the disperse phase for water (w) is

$$\frac{dM^j_{p,w}}{dt} = -\dot{m}^j_{evap} \tag{6}$$

where $M^j_{p,w}$ is the mass of water in the disperse phase or raffinate.

For maltodextrin (mx) the balance gives $\frac{dM^j_{p,mx}}{dt} = 0$ as this compound remains constant during the whole process.

Energy balance on a partition containing droplets or particles:

The result of applying the energy balance on a disperse phase p partition is

$$\frac{d}{dt}\left(M^j_{p,mx}\,\widehat{U}^j_{p,mx} + M^j_{p,w}\,\widehat{U}^j_{p,w}\right) = \dot{Q}^j - \dot{m}^j_{evap}\,\widehat{H}^j_{evap} \tag{7}$$

where $\widehat{U}^j_{p,n}$ is the internal energy of an n compound of the disperse phase, i.e., maltodextrin (mx) and water (w). Proceeding as for PS II, the final expression is

$$\frac{dT^j_p}{dt} = \frac{\dot{Q}^j - \dot{m}^j_{evap}\,\widehat{H}^j_{evap} - \widehat{U}^j_{p,w}\,\frac{dM^j_{p,w}}{dt}}{M^j_{p,mx}\left(T^j_p\,\frac{dC^j_{v\,p,mx}}{dT} + C^j_{v\,p,mx}\right) + M^j_{p,w}\left(T^j_p\,\frac{dC^j_{v\,p,w}}{dT} + C^j_{v\,p,w}\right)} \tag{8}$$

where T^j_p is the temperature of the disperse phase, and $C^j_{v\,p,n}$ the heat capacity of component n at a constant volume.

Mechanical energy balance:

A force balance on the droplet replaces the mechanical energy balance. This is intended to approximate the position of a group of particles under specific conditions. There are two physical limits for the trajectories, the border and the center of the tower. Situations in which droplets collapse at the center of the tower, or at the chamber walls, generating an agglomerate of maltodextrin or bigger droplets, are avoided. Newton's law is applied, producing a force balance as follows

$$\frac{d\mathbf{v}^j_p}{dt} = \mathbf{g} - \frac{\sum \mathbf{F}^j_p}{M^j_p} \tag{9}$$

The term $\mathbf{v}^j_p$ refers to droplet velocity as this is produced until the droplet reaches the bottom of the conical section of the chamber. This variable corresponds to a vector of three components, each represented by $v^j_{p,i}$, where i specifies the dimension. This force balance includes the axial (x), radial (r), and tangential (t) dimensions. The term $\mathbf{g}$ is the gravitational acceleration constant, M^j_p represents the mass of a single droplet, and the term $\mathbf{F}^j_p$ is an individual force experienced by the droplet. The forces that must be considered in spray drying are listed and explained in detail in Mujumdar (2015). In this work, the forces considered are the drag, buoyancy, and added mass forces symbolized as $\mathbf{F}^j_D$, $\mathbf{F}^j_B$, and $\mathbf{F}^j_A$, respectively. All vectors in this work, including forces, also consist of the three mentioned dimensions and maintain the same nomenclature (Table 1). Newton's law is therefore described as follows,

Table 1 Nomenclature

A_0	Discharge orifice area (m^2)
a_1, a_2, a_3	Drag coefficient parameters
A_a	Air core area (m^2)
A_p	Total inlet ports area (m^2)
a_w	Water activity
B	Spalding number
C_1, C_2	Air flow pattern parameters
C_D	Discharge coefficient
C_{drag}	Drag coefficient
C_{eq}	Equilibrium moisture parameter
C_p	Heat capacity at constant pressure (kJ/kg K)
C_v	Heat capacity at constant volume (kJ/kg K)
d_0	Discharge orifice diameter (m)
D	Diameter (m)
d_{air}	Diffusion coefficient of water vapor in air (m^2/s)
D_{eff}	Effective diffusion of vapor through the solid (m^2/s)
d_p	Diameter of nozzle ports (m)
D_s	Swirl-chamber diameter (m)
$\mathbf{F}_A$	Added mass force (N)
$\mathbf{F}_B$	Buoyancy force (N)
$\mathbf{F}_D$	Drag force (N)
F_{Num}	Flow number (m^2)
g	Gravity (m/s^2)
h	Heat transfer coefficient (kW/m^2 K)
$\hat{H}$	Evaporated water enthalpy (kJ/kg)
K	Atomizer constant
K_{eq}	Equilibrium moisture parameter
K_v	Velocity coefficient
k_y	Local mass transfer coefficient in gas phase (m/s)
l	Length or distance (m)
L_s	Parallel portion length of the swirl-chamber (m)
$\dot{m}$	Mass flow rate (kg/s)
M	Mass (kg)
MW	Molecular weight (kg/kmol)
n_p	Number of droplets per iteration
P	Pressure (Pa)
Pr	Prandtl number
$\dot{Q}$	Rate of heat flow (kW)
r	Radial distance traveled (m)
R	Gas constant (kJ/kmol K)
r_a	Final discharge orifice radius (m)
Re	Reynolds number
R_s	Swirl-chamber radius (m)
Sc	Schmidt number
Sh	Sherwood number
SMD	Sauter mean diameter (m)
T	Temperature (°C)
T'	Temperature (K)
t_f	Film thickness (m)
$\hat{U}$	Internal energy (kJ/kg)
U_h	Overall heat transfer coefficient (kW/m^2 °C)
v	Velocity (m/s)
W_{eq}	Equilibrium moisture parameter
x or y	Mass fraction (kg/kg_{mix})
X or Y	Solute-free basis mass fraction ($kg/kg_{solvent}$)
Greek letters	
ρ	Density (kg/m^3)
μ	Dynamic viscosity (kg/m s)
κ	Thermal conductivity (kW/m K)
λ	Specific heat of evaporation (kJ/kg)
σ	Stefan-Boltzmann constant (kW/m^2 K^4)
σ_p	Surface tension (N/m)
ϵ	Emissivity
χ	Areas ratio
θ	Mean spray cone half angle (°)
δ	Drop path per iteration (m)
Δ	Change
Subscripts	
0	Initial condition
a	Air
amb	Ambient
av	Average
$Atom$	Atomizer
cn	Constant
$conv$	Convection
cr	Critical
eq	Equilibrium
eqp	Equipment
$evap$	Evaporated
g	Gas
L	Lactose
mx	Maltodextrin
o	Orifice
p	Particles or liquid phase
r	Radial direction
rad	Radiation
rel	Relative
s	Sensible heat
sat	Saturation
t	Tangential direction
v	Vapor
w	Water
x	Axial direction

$$\frac{d\mathbf{v}_p^j}{dt} = \mathbf{g} - \frac{\mathbf{F}_D^j + \mathbf{F}_B^j + \mathbf{F}_A^j}{M_p^j} \tag{10}$$

Droplets trajectories are predicted by solving the following equation numerically,

$$\frac{d\mathbf{l}_p^j}{dt} = \mathbf{v}_p^j \tag{11}$$

It is worth mentioning that the axial component of the droplet trajectory, $l_{p,x}^j$, is related to the numerical method and the delimitation of the air partitions. The distance travelled by a group of droplets in an iteration of time delimits the air partition. Furthermore, the air phase transfers heat to the droplets, but it also losses heat through the tower walls. The path of the droplets in each partition was used to calculate the heat losses, which is reflected in the parameter δ_p^j of the constitutive equation of heat transfer rate to the ambient $\dot{Q}_{amb}^j$.

Variables, structural parameters, and structural constants

The basic structure of the model is composed by Eqs. (2), (4), (6), (8), and (10). These equations provide enough information to fulfill the model objective. Table 2 presents the variables, parameters, and constants of the model.

Constitutive, assessment equations, and constants definition

Typically, constitutive and assessment equations are algebraic and either based on accepted laws and theories or from arbitrary black-box structures. The most important constitutive equations are described in detail below. The definition of symbols and units are shown in Table 1. Due to space limitation the rest of the model parameter deduction is not presented here. However, Table 3 lists all of the model's extended structure, based on constitutive and assessment equations and their origins. The selection of the empirical equations used in the model was guided by the range of applicability and compatibility of each expression. The reader may review the given literature.

Nozzle equations

The empirical equations that describe and fit the nozzle behavior are shown in this section. Their terms are described only briefly as this is not the objective of this work.

Average velocity (Lefebvre and McDonell 2017)

$$v_{av} = K_v \sqrt{\frac{2\left(P_{p,in} - P_{p,out}\right)}{\rho_{p,0}}} \tag{12}$$

$$K_v = 0.00367\,(K)^{0.29}\left(\frac{P_L\,\rho_{p,0}}{\mu_p}\right)^{0.2} \tag{13}$$

$$K = \frac{A_p}{D_s\,d_0} \tag{14}$$

Liquid flow rate of raffinate (Lefebvre and McDonell 2017)

$$\dot{m}_{p,0} = 0.802\,C_D\,\frac{A_0\,\rho_{p,0}\,v_{av}}{K_v} \tag{15}$$

$$C_D = 0.45\left(\frac{d_0\,\rho_{p,0}\,v_{av}}{\mu_p}\right)^{-0.02}\left(\frac{l_0}{d_0}\right)^{-0.03}\left(\frac{L_s}{D_s}\right)^{0.05}\left(\frac{A_p}{D_s\,d_0}\right)^{0.52}\left(\frac{D_s}{d_0}\right)^{0.23} \tag{16}$$

$$A_0 = \pi\left(\frac{d_0}{2}\right)^2 \tag{17}$$

Cone angle (Lefebvre and McDonell 2017)

The mean spray cone half-angle is the middle of the cone angle shown in Fig. 2. The following expression corresponds to an empirical equation for this parameter in degrees

$$\theta = 8.078\,K^{-0.39}\,d_0^{1.13}\,\mu_p^{-0.9}\,P_L^{0.39} \tag{18}$$

Table 2 Model variables, structural parameters, and constants

	Classification	Total
Variables	$M_{p,w}, M_{g,w}, T_g, T_p, \mathbf{v}_p, \mathbf{l}_p$	10
Parameters	$\dot{m}_{evap}, \widehat{H}_{evap}, \dot{Q}, \dot{Q}_s, \dot{Q}_{amb}, \widehat{U}_{g,w}, M_{g,a}, C_{vg,a}, C_{vg,w}, \widehat{U}_{p,w}, M_{p,mx}, C_{vp,mx}, \mathbf{F}_D, \mathbf{F}_B, \mathbf{F}_A, M_p, v_{av}, K_v, P_{p,in}, P_{p,out}, \rho_{p,0}, \mu_p, A_p, D_s, d_0, \dot{m}_{p,0}, A_0, C_D, l_0, L_s, \theta, K, v_{p,0}, v_{p,0,x}, v_{p,0,r}, v_{p,0,t}, A_a, \chi, r_a, t_f, F_{Num}, SMD, \sigma_p, \rho_{g,0}, n_p, D_p, P, D_{eff}, f, k_y, P_{g,w}, P_{sat,p}, X_{cr}, \rho_{p,w}, \widehat{H}_{evap}, C_{pp,w}, \lambda, \rho_L, \rho_w, h, h_{conv}, h_{rad}, Re, \rho_{g,a}, \mu_{g,a}, Pr, C_{pg,a}, k, B, C_{drag}, \mathbf{v}_g, v_{g,r}, v_{g,t}, Sh, Sc, D_{air}, X_{eq}, C_{eq}, K_{eq}, W_{eq}, a_w$	89
Constants	$g, MW_w, R, \sigma, \epsilon$	5

Table 3 Constitutive and assessment equations for structural parameters

Description	Equation	Units	Refs.
Initial density of the liquid	$\rho_{p,0} = \frac{100000}{(1-x_w)\left(\frac{1}{\rho_L}-\frac{1}{\rho_w}\right)+\frac{100}{\rho_w}}$	[kg/m^3]	Sivarajalingam (2009)
Mass fraction of water in the liquid	$x_w = Datum$	[kg/kg]	Sivarajalingam (2009)
Lactose density	$\rho_L = 1.635 - 0.0026\,T_p + 2\cdot 10^{-5}\,T_p^2$	[kg/m^3]	Sivarajalingam (2009)
Liquid water density	$\rho_w = 1.0020825 - 1.14\cdot 10^{-4}\,T_p - 3.325\cdot 10^{-6}\,T_p^2$	[kg/m^3]	Sivarajalingam (2009)
Evaporated water enthalpy	$\widehat{H}_{evap} = \int_{T_{Ref}}^{T_p} C_{p\,p,w}\,dT + \lambda(T_p)$	[kJ/kg]	Felder et al. (2020)
Heat capacity of liquid water	$C_{p,pw} = 4.18$	[kJ/(kg K)]	Felder et al. (2020)
Specific heat of evaporation	$\lambda = 3.15\cdot 10^3 - 2.38\,T_p'$	[kJ/kg]	Dalmaz et al. (2007)
Heat flow rate from the gas to the liquid	$\dot{Q} = n_p\,\pi\,D_p^2\,h\left(T_g - T_p\right)$	[kW]	Felder et al. (2020)
Number of droplets per iteration	$n_p = \frac{\dot{m}_p}{\frac{\pi}{6}\,SMD^3\,\rho_{p,0}}$	[–]	Own development
Liquid flow rate	$\dot{m}_p = \frac{C_D\,A_0\,\rho_{p,0}\,v_{av}}{K_v}$	[kg/s]	Lefebvre and McDonell (2017)
Liquid flow rate parameters	$C_D\,A_0,\ v_{av},\ K_v$: Please, see "Nozzle equations" section	[–]	Lefebvre and McDonell (2017)
Sauter mean diameter	SMD : Please, see "Nozzle equations" section	[m]	Lefebvre and McDonell (2017)
Droplet diameter	D_p : Please, see "Mass transfer rate and droplet diameter" section	[m]	Sivarajalingam (2009)
Overall heat transfer coefficient	$h = h_{rad} + h_{conv}$	[kW/(m^2 K)]	Mezhericher et al. (2008a)
Radiant heat transfer coefficient	$h_{rad} = \sigma\,\epsilon\left(\frac{T_g^4 - T_p^4}{T_g - T_p}\right)$	[kW/(m^2 K)]	Mezhericher et al. (2008a)
Stefan–Boltzman constant	$\sigma = Datum$	[kW/(m^2 K^4)]	Felder et al. (2020)
Emissivity	$\epsilon = Datum$	[−]	Dalmaz et al. (2007)
Convective heat transfer coefficient	$h_{conv} = \frac{\left(2+0.6\,Re^{1/2}\,Pr^{1/3}\right)\kappa}{D_p\,(1+B)^{0.7}}$	[kW/(m^2 K)]	Mezhericher et al. (2008a)
Reynolds number	$Re = \frac{\rho_{g,a}\,D_p\,v_{rel}}{\mu_{g,a}}$	[–]	Levi-Hevroni et al. (1995)
Air density	$\rho_{g,a} = 1.293\left(\frac{273.15}{T_g}\right)$	[kg/m^3]	Dalmaz et al. (2007)
Relative velocity of droplets and the gas	$v_{rel} = \sqrt{\left(v_{p,x} - v_{g,x}\right)^2 + \left(v_{p,r} - v_{g,r}\right)^2 + \left(v_{p,t} - v_{g,t}\right)^2}$	[m/s]	Own development
Axial gas velocity[a]	$v_{g,x} = C_1\left(1 - \frac{r-0.1}{D_{eqp}}\right)^{20} + 1$	[m/s]	Shabde and Hoo (2008)
Radial gas velocity[a]	if $l_p \le 0.1\,m$: $v_{g,r} = C_2$, else: $v_{g,r} = 0$	[m/s]	(Kieviet 1997)
Tangential gas velocity[a]	$v_{g,t} = C_3\left(\frac{D_{eqp}-r}{D_{eqp}}\right)^{0.5}$	[m/s]	Shabde and Hoo (2008)
Gas velocity parameters[a]	$C_1, C_2, C_3 = Identified\ parameters$	[–]	Kieviet (1997)
Spray drying chamber diameter	if $l_p \le 2.0\,m$: $D_{eqp} = 2.2$, else: $D_{eqp} = \frac{2\sqrt{3}}{3}\left(3.9 - l_p\right)$	[m]	Own development
Air dynamic viscosity	$\mu_{g,a} = 1.72\cdot 10^{-5} + 4.568\cdot 10^{-8}\,T_g$	[kg/(m s)]	Dalmaz et al. (2007)
Prandlt number	$Pr = \frac{\mu_{g,a}\,C_{p\,g,a}}{\kappa}$	[–]	Levi-Hevroni et al. (1995)
Air heat capacity	$C_{p\,g,a} = A_{cn,1} + B_{cn,1}\,T_g' + C_{cn,1}\,T_g'^2 - D_{cn,1}\,T_g'^3$	[kJ/(kg K)]	Dalmaz et al. (2007)
Heat capacity coefficients	$A_{cn,1},\ B_{cn,1},\ C_{cn,1},\ D_{cn,1} = Data$	[–]	Dalmaz et al. (2007)
Air thermal conductivity	$\kappa = 1.731\left(0.014 + 4.296\cdot 10^{-8}\,T_g\right)$	[kW/(m K)]	Dalmaz et al. (2007)
Spalding number	$B = \frac{C_{p,gw}\left(T_g - T_p\right)}{\lambda}$	-[–]	Mezhericher et al. (2008a)
Sensible heat to the evaporated vapors	$\dot{Q}_s = \int_{T_p}^{T_g} C_{p\,g,w}\,dT$	[kW]	Felder et al. (2020)
Heat capacity of water vapor at constant pressure	$C_{p\,g,w} = A_{cn,2} - B_{cn,2}\,T_g + C_{cn,2}\,T_g^2 - D_{cn,2}\,T_g^3$	[kJ/(kg K)]	Dalmaz et al. (2007)
Heat capacity coefficients	$A_{cn,2},\ B_{cn,2},\ C_{cn,2},\ D_{cn,2} = Data$	[–]	Dalmaz et al. (2007)
Heat transfer rate to the ambient	$\dot{Q}_{amb} = \pi\,D_{eqp}\,\delta_p\,U_h\,(T_g - T_{amb})$	[kW]	Felder et al. (2020)
Overall heat transfer coefficient	$U_h = Identified\ parameter$	[kW/(m^2 °C)]	Own development
Internal energy of water vapor	$\widehat{U}_{g,w} = \int_{T_{Ref}}^{T_g} C_{v\,g,w}\,dT$	[kJ/kg]	Felder et al. (2020)

Table 3 (continued)

Description	Equation	Units	Refs.
Heat capacity of water vapor at constant volume	$C_{v\,g,w} = C_{p\,g,w} - R$	[kJ/(kg K)]	Felder et al. (2020)
Internal energy of liquid water	$\widehat{U}_{p,w} = \int_{T_{Ref}}^{T_p} C_{v\,p,w}\, dT$	[kJ/kg]	Felder et al. (2020)
Heat capacity coefficient of liquid water at constant volume	$C_{v\,p,w} = C_{p\,p,w} - R$	[kJ/(kg K)]	Felder et al. (2020)
Drag force vector	$\mathbf{F}_D = \frac{\pi}{8}\,\rho_g\, D_p^2\, C_{drag}\, \lvert\mathbf{v}_g - \mathbf{v}_p\rvert(\mathbf{v}_g - \mathbf{v}_p)$	[N]	Mujumdar (2015)
Drag coefficient	$C_{drag} = \frac{a_1}{Re} + \frac{a_2}{Re^2} + a_3$	[–]	Morsi and Alexander (1972)
Drag coefficient parameters	$a_1, a_2, a_3 = Datum$	[–]	Morsi and Alexander (1972)
Added mass force vector	$\mathbf{F}_A = \rho_g\, \frac{\pi\, D_p^3}{12}\, \frac{d}{dt}\left(\mathbf{v}_g - \mathbf{v}_p\right)$	[N]	Mujumdar (2015)
Buoyancy force vector	$\mathbf{F}_B = -\frac{\pi}{8}\,\rho_g\, D_p^3\, \mathbf{g}$	[N]	Mujumdar (2015)
Mass per droplet	$M_p = \left(1 - x_{w,0}\right)\rho_{p,0}\, \frac{\pi\, SMD^3}{6}$	[kg]	Own development
Dynamic viscosity of the liquid	$\mu_p = Datum$	[kg/(m s)]	Sivarajalingam (2009)
Effective diffusivity of water through the solid	$D_{eff} = Datum$	[m^2/s]	Sivarajalingam (2009)
Local mass transfer coefficient in the gas phase	$k_y = Sh\, \frac{D_{air}}{D_p}$	[m/s]	Sivarajalingam (2009)
Sherwood number	$Sh = 2 + Re^{0.5} + Sc^{0.3333}$	[–]	Sivarajalingam (2009)
Schmidt number	$Sc = \frac{\mu_{g,a}}{\rho_{g,a}\, D_{air}}$	[–]	Sivarajalingam (2009)
Diffusion coefficient of water vapor in air	$D_{air} = 2.302 \cdot 10^{-5}\, \frac{P_{cn}}{P} \left(\frac{T'_g}{T_{cn}}\right)^{1.81}$	[m^2/s]	Mezhericher et al. (2008a)
Pressure constant of the diffusion coefficient	$P_{cn} = 0.98 \cdot 10^5$	[Pa]	Mezhericher et al. (2008a)
Temperature constant of the diffusion coefficient	$T_{cn} = 256$	[K]	Mezhericher et al. (2008a)
Water mass fraction in solvent-free basis	$X_{eq} = \frac{C_{eq}\, K_{eq}\, W_{eq}\, a_w}{(1-K_{eq}\, a_w)(1-K_{eq}\, a_w + C_{eq}\, K_{eq}\, a_w)}$	[kg/kg]	Sivarajalingam (2009)
Equilibrium moisture content parameter	$C_{eq} = 0.04\, exp\left(\frac{1257.14}{T'_p}\right)$	[–]	Sivarajalingam (2009)
Equilibrium moisture content parameter	$K_{eq} = 0.65\, exp\left(\frac{144.57}{T'_p}\right)$	[–]	Sivarajalingam (2009)
Equilibrium moisture content parameter	$W_{eq} = 0.05\, exp\left(\frac{-99.27}{T'_p}\right)$	[–]	Sivarajalingam (2009)
Water activity	$a_w = \frac{P_v}{P_{sat,p}}$	[–]	Sivarajalingam (2009)
Liquid saturation pressure	$P_{sat,p} = \left(\frac{101325}{760}\right) \cdot 10^{7.95581 - \frac{1668.210}{T_p + 228}}$	[Pa]	Richard M. Felder (2018)
Critical moisture content in a solvent-free basis	$X_{cr} = Datum$	[kg/kg]	Sivarajalingam (2009)

[a]Gas velocity equations presented in the appendix of Shabde and Hoo (2008) were modified to adjust velocity profiles reported in Kieviet (1997)

Velocity components (Lefebvre and McDonell 2017)

$$v_{p,0} = \sqrt{v_{p,0,x}^2 + v_{p,0,r}^2 + v_{p,0,t}^2} \tag{19}$$

$$v_{p,0,x} = \frac{\dot{m}_{p,0}}{\rho_{p,0}\left(A_0 - A_a\right)} \tag{20}$$

$$v_{p,0,r} = \frac{\dot{m}_{p,0}\, R_s}{\rho_{p,0}\, A_p\, r_a} \tag{21}$$

$$v_{p,0,t} = v_{p,0,x}\, tan(\theta) \tag{22}$$

$$A_a = A_0\, \chi \tag{23}$$

$$r_a = \sqrt{\frac{A_a}{\pi}} \tag{24}$$

$$\chi = \left(\frac{d_0 - 2\,t_f}{d_0}\right)^2 \tag{25}$$

$$t_f = 3.66\left(\frac{d_0\,F_{Num}\,\mu_p}{\sqrt{P_L\,\rho_{p,0}}}\right)^{0.25} \tag{26}$$

$$F_{Num} = \frac{\dot{m}_{p,0}}{\sqrt{\rho_{p,0}\,P_L}} \tag{27}$$

Initial droplet diameter (Milan 2014)

$$SMD = 1.1012 \cdot 2.25\left(\sigma_p^{0.25}\right)\left(\mu_p^{0.25}\right)\left(\dot{m}_{p,0}^{0.22}\right)\left(P^{-0.5}\right)\left(\rho_{g,0}^{-0.25}\right) \tag{28}$$

Mass transfer rate and droplet diameter

A wide variety of equations to determine the mathematical description of the mass transfer rate have been proposed (Petersen 2016; Sivarajalingam 2009; Kieviet 1997; Birchal et al. 2006). The apparent reason for this is that each equation is applied depending on the compounds involved. Still, it is common to find different mass transfer rate equations applied to the same compounds (Petersen 2016; Sivarajalingam 2009), and the same equations to different compounds (Petersen 2016; Birchal et al. 2006). In this work, the mass transfer rate proposed by Clement et al. (1991) and used in Sivarajalingam (2009) and Birchal et al. (2006) is implemented in the drying model. It is applied for spherical particles and is a good approach to the drying mechanisms because of its *f* factor, which changes when the critical moisture content is reached. The mass flow rate equation and the *f* factor are described as follows

$$\dot{m}_{evap} = n_p\,\pi\,D_p\,\frac{P\,MW_w}{RT_p}\,\frac{2\,D_{eff}}{\left(f + \frac{2\,D_{eff}}{k_y\,D_p}\right)}\,\ln\left(\frac{P - P_{g,w}}{P - P_{sat,p}(T_p)}\right) \tag{29}$$

$$f = 0 \quad if \;\; X_p > X_{cr}$$
$$f = \left(\frac{X_p - X_{eq}}{X_{cr} - X_{eq}}\right)^{-\frac{1}{3}} - 1 \quad if \;\; X_p < X_{cr} \tag{30}$$

Regarding Eq. (30), the X_p is the moisture content fraction of a droplet, the X_{eq} is the equilibrium moisture fraction, and the X_{cr} is the critical moisture fraction of the particle. It is noted that all these concentrations are moisture-free basis fractions. In addition, the droplet diameter is given by Sivarajalingam (2009)

$$D_p = \left(D_{p,0}^3 - \frac{6\,M_{p,mx}\left(X_{p,0} - X_p\right)}{\pi\,n_p\,\rho_{p,w}}\right)^{\frac{1}{3}} \tag{31}$$

$$D_p = \left(D_{p,0}^3 - \frac{6\,M_{p,mx}\left(X_{p,0} - X_{cr}\right)}{\pi\,n_p\,\rho_{p,w}}\right)^{\frac{1}{3}} \tag{32}$$

Table 3 presents the constitutive or assessment equation for each of the structural parameters that appear in the basic model structure.

Degrees of freedom analysis

A solvable set of equations is obtained when the degrees of freedom of that set are zero. In this case, the number of equations is equal to the sum of variables and parameters. The model contains 10 variables and 89 parameters; i.e., 99 unknowns. In addition, there are 99 equations in total forming the basic model structure and the extended model structure. Therefore, the model is solvable as $99 - 99 = 0$. The high number of unknowns is not indicative of model complexity because from these large number of unknowns, only four have to be identified based on experimental data, the rest have a given constitutive or assessment equation to determine the unknown value. Nevertheless, meticulous measurements or an accurate steady-state model as CFD were used to identify all the model parameters.

Model simulation and contrasting

This final section describes the computational model, including the proposed numerical methodology to solve it. The section ends with the model contrasting.

Computational model and numerical methodology

This work implements a numerical methodology to represent the dynamic behavior of the system without solving complex systems of partial equations, the use of excessive simplifications as the CSTR assumption, and keeping the accumulation term of the balances which is omitted with the PFR simplification. These explicit terms allow the direct implementation of control methods. This methodology is a sequential solution of phases and partitions, which are solved as semi-batch processes. Several features of spray drying allow its implementation. Firstly, the droplets are faster than the air during the whole operation. This velocity

gap is more noticeable at the top of the tower. Secondly, the continuous flow of both phases can be discretized as semi-batches. If the time differential of these sequential calculi is low enough, their superposition can represent the dynamics of the system. In this case, these calculi are solved with the Euler method.

A scheme of the sequential process is shown in Fig. 4. The methodology begins when the initial conditions and the gas phase partitions are established. These partitions may be deemed as temporary stagnant sections of gas. Moreover, the disperse phase is also separated into partitions. When the nozzle produces a certain number of droplets, the first semi-batch is formed and falls throughout the tower interacting only with the stagnant air partitions. That is possible due to the mentioned velocity gap. During the semi-batch movement, its time at a specific partition is stored. That is useful for the subsequent semi-batch, which takes place one time differential after the current semi-batch, and overlapping to the semi-batch information for the dynamic solution. Eventually, the current semi-batch reaches the tower bottom, and the air shifts its position. This shift corresponds to the average gas phase velocity and is indicated as dashed arrows in Fig. 4. Finally, dry air enters the chamber before a second semi-batch of droplets retaking the sequence. The range of the air phase velocity for a system with similar dimensions and flow may vary between 8 $\frac{m}{s}$ in the center of the tower to almost 0 $\frac{m}{s}$ towards the wall. In contrast, the initial velocity of the droplets at the nozzle may be higher than 50 $\frac{m}{s}$ when balances are essential for crust formation and then get slower along their path (Kieviet 1997). After the crust formation and the velocity reduction of the droplets at a slightly higher velocity than the air, this numerical methodology becomes less accurate in overestimating the velocity of droplets due to the main assumptions.

It is worth mentioning that the position shift in this methodology is closely related to gas velocity average requiring a constant correction to ensure that mass and energy balances are satisfied. The correction may be implemented as a factor close to 1 that affects the average gas velocity. If the gas accumulates in the system generating a gas excess, the velocity will increase by a small factor. In the opposite case, the velocity decreases. The mass balances in a steady state are used as references for the correction. Furthermore, the average gas velocity and the profiles shown in Table 3 are decoupled. The former affects the balances and is corrected continuously, while the latter is used to approximate the trajectory of the particles from previous CFD profiles.

Storing data is also relevant in this process because each semi-batch of droplets is solved individually during their complete trajectory. The semi-batch solutions at specific times and distances are overlapped to form a macroscopic solution. Since storing that significant amount of information represents a high computational cost, storing data at specific semi-batches is recommended. However, the process requires temporary storage of each phase solution, especially the air conditions modified by each semi-batch of droplets. Regarding computational cost, the step in which information is stored and used is the most critical. The algorithm explained in this section is detailed with all the additional steps in Fig. 5. The numerical methodology developed in this work can be used in co-current and counter-current models. Nevertheless, it is impossible to use the current version in mixed-flow spray dryers without significant adjustments.

Model contrasting

Since there is no data on the dynamic behavior of a spray drying tower for various axial positions, two different data sets have to be used to contrast the proposed model. The first one, taken from Kieviet (1997), contains information related to the steady states reached in various axial positions of a co-current spray drying tower. The second one, taken from Petersen (2016), provides information on the dynamic behavior of the outlet temperature of a similar system; i.e., a mixed-flow spray dryer.

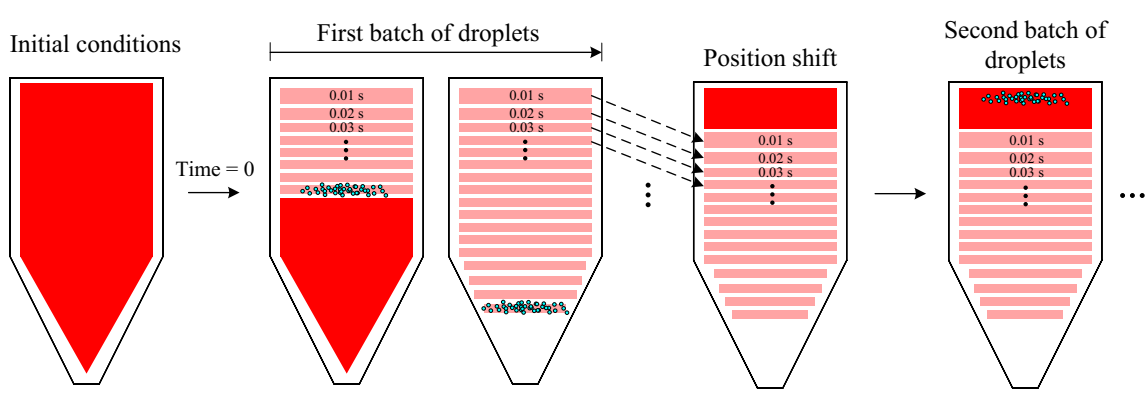


Fig. 4 Numerical methodology scheme

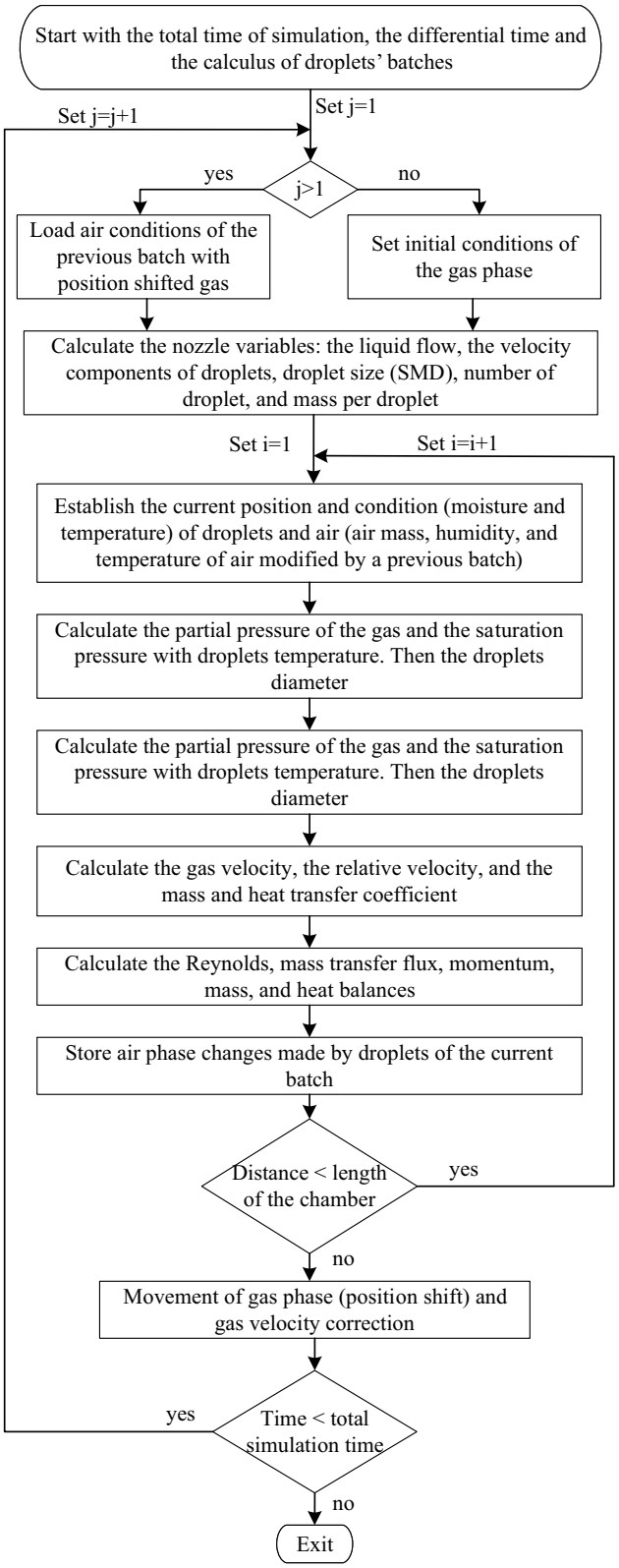


Fig. 5 Proposed numerical methodology algorithm to solve the model

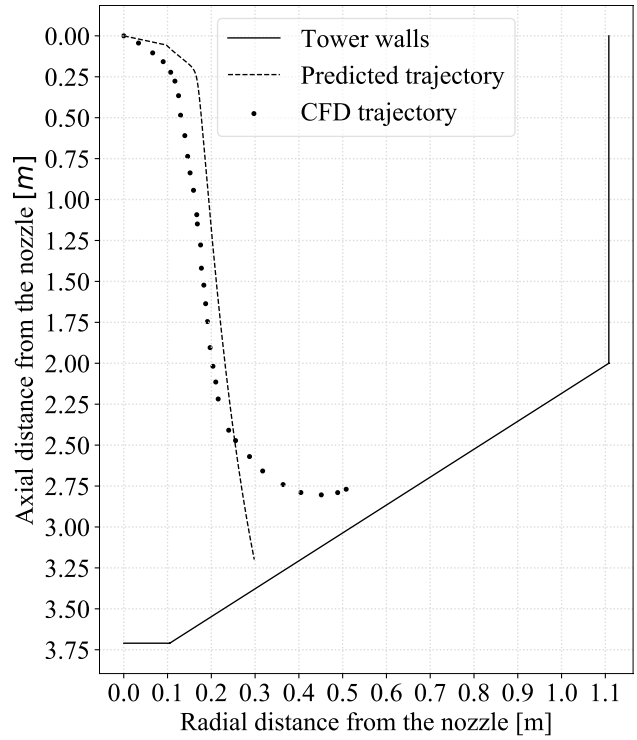
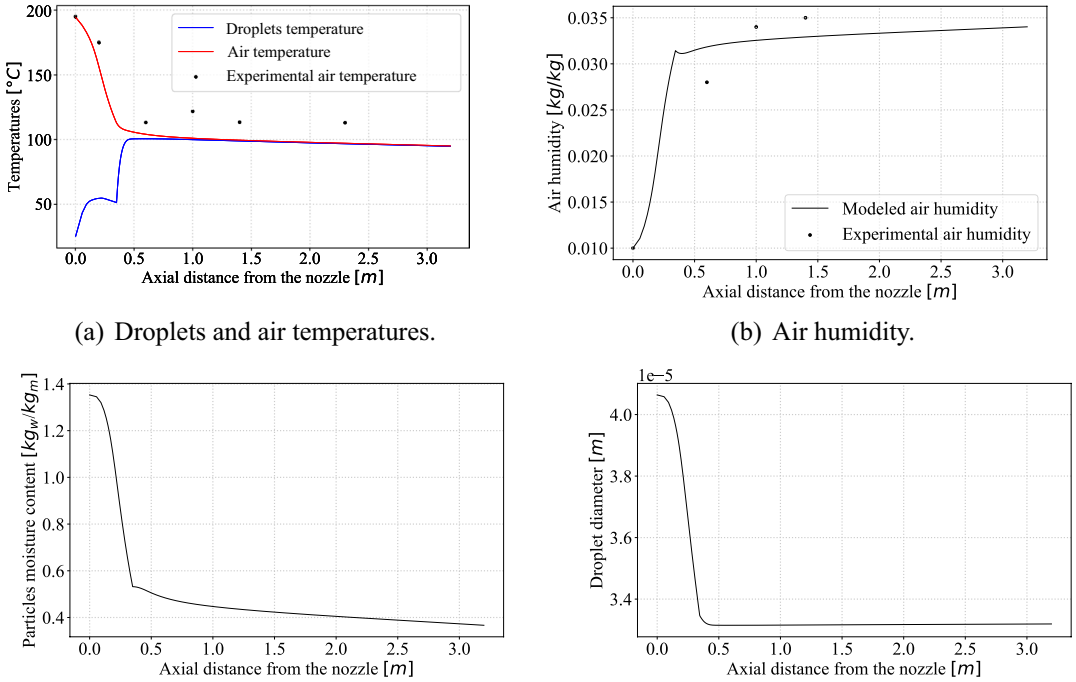


Fig. 6 Predicted particles trajectory (dashed line) and comparison with CFD results (points) reported in (Kieviet 1997)

The experimental and predicted values for the radial particle trajectory, phase temperatures, air humidity, moisture content, and droplet diameter are sketched axially along the drying tower in Figs. 6 and 7. These figures do not contain dynamic information. This means that the properties mentioned are predicted from the nozzle to the bottom of the tower at a specific time or at a steady state. In this case, an airflow of 1150 kg/h was used to dry a liquid flow of 66 kg/h. Moreover, the continuous phase temperature and humidity are compared with experimental data from Kieviet (1997) in a steady state. The mentioned work uses CFD simulation to generate some data contained in the final report of this work. This data are used here to identify some parameter of the current model. In the same way, equipment dimensions and conditions of Kieviet (1997) are used in the proposed model.

Regarding the comparison between model predictions and the real dynamic behavior, the dynamics of the outlet temperature corresponds to another data set and a different system. The outlet temperature is compared with experimental data from Petersen (2016). The relevance of this comparison resides in the steady-state values reached after dynamic behaviors and the velocity in which the system reaches these values. Due to the comparison of units with different configurations, their dynamic behavior is discussed in the following section. The dimensions and conditions of both units



(a) Droplets and air temperatures.

(b) Air humidity.

(c) Particles moisture content

(d) Droplet diameter.

Fig. 7 Predicted spray dryer steady state compared with averaged experimental data (Kieviet 1997)

are adjusted in the proposed model, but the configuration of each unit remains at stated in the cited work.

Results and discussion

As mentioned earlier, Fig. 6 presents the output of a comparison between particle trajectory and CFD results taken directly from (Kieviet 1997). The trajectory prediction aims to approximate the position of droplet batches and the moment when the drying finishes. In this case, the trajectory is part of the complete hollow cone formed by the nozzle and fits the CFD results from the nozzle to distances close to the tower's edges. Close to the walls, the air pattern changes and is predicted by the CFD but not by the current model. These changes lead to the small particles being dragged through the walls. However, the deviation of the particle does not make a difference in mass and heat transfer because of the co-current nature of the equipment. Precisely, before the mentioned deviation the heat and mass transfer effects are very slow. Therefore, the final effect of this mismatch over model precision is not important, as it is evident from next figures.

Droplets and air temperatures are presented in Fig. 7a. It is noted that experimental points are obtained through the average of various horizontal points in a co-current spray dryer with a liquid flow rate of 50 L/h and a gas flow rate of 1150 kg/h. The predicted values were less than the actual gas temperature in all cases, but the behavior trend is clear. The abrupt changes at the top of the tower fitted better, and after 0.7 m from the nozzle, the prediction deviated from the experimental temperatures. However, this deviation does not affect the model prediction of air humidity. Mentioned deviation may be due to unpredicted air returns from the bottom through the walls, reducing the ambient heat loss. An air fraction can return from the bottom of the tower to the top, recycling near the nozzle, which also reduces temperatures from the beginning of the process. This effect may also affect the prediction of air humidity. It is important to highlight that for the model uses in control this mismatch is not significant and therefore, it can be easily rejected by the controller. In addition, these residual air returns imply an increase in the complexity of the model. Figure 7b shows the rising experimental values of the water-free basis mass fraction of air humidity over the modeled line with the evident good agreement of the model.

The moisture content of the particles and droplet diameter are predicted in Fig. 7c, d but not compared with experimental data as it is not possible to measure those values at various axial positions in the tower. In this case, the prediction of the droplet diameter does not include how heat expands the formed crust. Following crust formation, droplet diameter remains constant. This effect should be considered when air temperature after crust formation exceeds

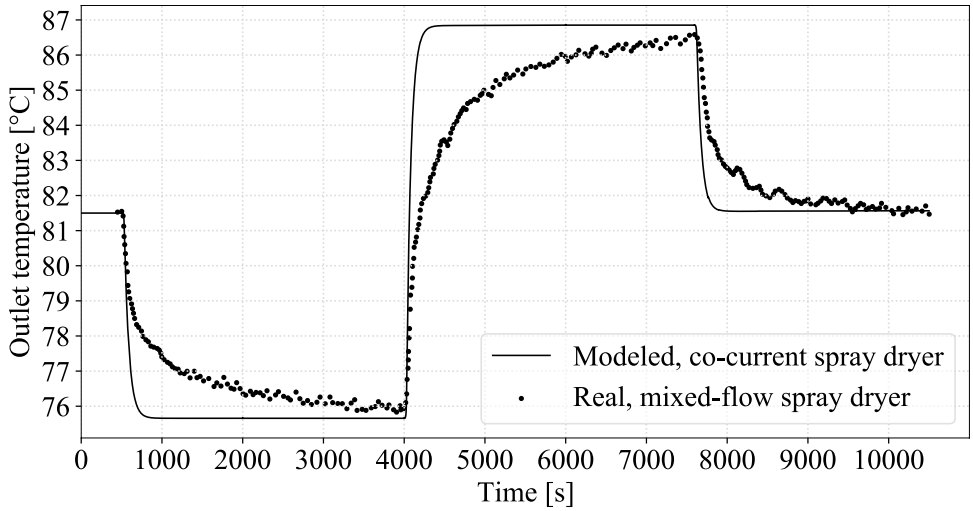


Fig. 8 Dynamic behavior comparison (Petersen 2016)

the boiling temperature. It is worth mentioning that all the steady state figures show the abrupt changes when the critical moisture is reached, and the crust is formed before the 0.5 m of the tower as expected. The proposed model is able to consider, with addition equations, the crust formation phenomena. This option is useful for particle with special crust characteristics.

Regarding the outlet temperature dynamics presented in Fig. 8, a faster behavior in simulation was expected due to the co-current configuration. In a mixed-flow spray dryer, the drying air enters the top of the chamber and travels throughout the whole tower until it leaves at the top, close to the inlet. That configuration promotes air mixing, a long residence time, and as a result a slower dynamic behavior. Conversely, the short residence time of the drying air in a co-current system makes it possible to reach the same steady states more quickly, which could be predicted through the proposed model. This comparison exposes the capability of the phenomenological based semi-physical models to capture the dynamic behavior of the process by using the conservation principle.

Conclusions

Spray drying systems are complex regardless of their configurations due to the characteristics of phases interaction. To use the Phenomenological Based Semi-physical Modeling (PBSM) approach provides the proposed model with all the advantages of this family of models, basic structure unity, model modularity, combination of level of detail and level of specification, and parameter interpretability. Using some of these properties, the model can be simplified or expanded to include new features of the process, to the point that even the constitutive or assessment equations could be changed to describe the drying of other substances. The proposed numerical methodology to solve the model reduces both complexity and information losses, which occur when the common simplifications are applied. It is worth mentioning that this numerical methodology can also be extended to counter-current spray drying with the same advantages. However, it is not convenient to apply it to mixed-flow spray dryers.

The proposed model differs from a classic PFR model as it can predict dynamic changes even with a dispersed phase with variable velocity. It is able to predict composition, temperature and velocity of both phases, continuous and disperse, at axial points in the tower. Despite the simplifications on the continuous phase, this model is helpful to applications requiring on-line simulations as process control. Thus, the model can be used for the analysis, optimization, and model based control of the spray drying tower. Obviously, complementary work should be conducted before using this model for any of the mentioned tasks in process engineering. Additional efforts are required to improve the accuracy of the model without increasing the simulation time. Different approaches are helpful for these purposes as novel numerical methodologies and creative representations of the system as modeling by zones.

Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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